AI & ML by Aditya Jain sir --> Lecture 3

Recap

```
#variables

age = 20
print(age)
```

```
20
```

```
print(Age)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_643/2955847039.py in <cell line: 0>()
----> 1 print(Age)

NameError: name 'Age' is not defined
```

Next steps: Explain error

```
print("age")
```

```
age
```

```
#datatype
type(10)
```

```
int
```

```
type(5.4)
```

```
float
```

```
type("AJ")
```

```
str
```

```
type(True)
```

```
bool
```

```
#typecasting
type("4.5")
```

```
str
```

```
type(float("4.5"))
```

```
float
```

```
a = input()
```

```
6
```

```
type(a)
```

```
str
```

```
a = int(input())
```

```
5
```

```
type(a)
```

```
int
```

```
len = float(input())
```

```
5.7
```

```
type(len)
```

```
float
```

```
len = input()
```

```
6.7
```

```
type(len)
```

```
str
```

HW

```
What does the following code do?

name = input("Enter your name:")

A. Prints the name

B. Takes input from the user

C. Deletes the name

D. Creates an error

Ans: B
```

```
What data type is returned by input()?

A. int

B. float

C. bool

D. str

Ans: D
```

```
Which function converts a string to an integer?

A.

str()

B.

float()

C.

int()

D.

bool()

Ans: C
```

```
num = int("50")

print(type(num))
```

```
A.

<class 'str'>

B.

<class 'int'>

C.

<class 'float'>

D. Error


Ans: B
```

```
num = int("50")

print(type(num))
```

```
<class 'int'>
```

```
type(num)
```

```
int
```

```
age = 25

age = 30

print(age)

A.

25

B.

30

C.

25 30

D. Error


Ans: B
```

```
# What will be the output?

name = "Rahul"

print("Hello")
print(name)
```

```
Hello
Rahul
```

Arithmatic Operators

```
#Arithmatic Operators
print(5+4)
```

```
9
```

```
print(4-8)
```

```
-4
```

```
print(10*3)
```

```
30
```

```
print(20/5)
```

```
4.0
```

```
print(20/6)
```

```
3.3333333333333335
```

```
#floored division operator
print(20//6)
```

```
3
```

```
print(10//5)
```

```
2
```

```
print(10/5)
```

```
2.0
```

```
# Modulus --> gives remainder --> %

print(10%3)
```

```
1
```

```
print(10%6)
```

```
4
```

```
print(10%2)
```

```
0
```

```
# power --> **

print(2**3)
```

```
8
```

```
print(3**2)
```

```
9
```

```
#Calculator -->
# take two int as input from the user, and print +,-,*,/,//,**,%
```

```
a = int(input())
b = int(input())

print("Addition: ", a+b)
print("Subtraction: ", a-b)
print("Multiplication: ", a*b)
print("Division: ", a/b)
print("Floored Division: ", a//b)
print("Modulus: ", a%b)
print("Power: ", a**b)
```

```
10
4
Addition:  14
Subtraction:  6
Multiplication:  40
Division:  2.5
Floored Division:  2
```

```
Modulus:  2
Power:  10000
```

Assignment Operators

```
x = 10
print(x)
```

```
10
```

```
x = 10
x += 2 # x = x + 2
print(x)
```

```
12
```

```
x = 10
x -= 4
print(x)
```

```
6
```

```
x = 10
x *= 2
print(x)
```

```
20
```

```
x = 10
x /= 2
print(x)
```

```
5.0
```

```
x = 10
x //= 2
print(x)
```

```
5
```

```
x = 10
x **= 2
print(x)
```

```
100
```

Comparison Operators

```
10>5
```

```
True
```

```
4 < 3
```

```
False
```

```
10 >= 7
```

```
True
```

```
5 <= 5
```

```
True
```

```
5 < 5
```

```
False
```

```
5 == 5
```

```
True
```

```
a = 5
```

```
6 == 5
```

```
False
```

```
6 != 5
```

```
True
```

```
7 != 7
```

```
False
```

Logical Operators

```
# and , or , not
```

```
4 > 3
```

```
True
```

```
5 > 2
```

```
True
```

```
(4 > 3) and (5 > 2)
```

```
True
```

```
(4 > 3) and (5 < 2)
```

```
False
```

```
(4 < 3) and (5 < 2)
```

```
False
```

```
(4 > 3) or (5 > 2)
```

```
True
```

```
(4 > 3) or (5 < 2)
```

```
True
```

```
(4 < 3) or (5 < 2)
```

```
False
```

```
# not  --> toggle

# True --> False
# False --> True

(4 > 7)
```

```
False
```

```
not(4 > 7)
```

```
True
```

```
not ( 4 == 5)
```

```
True
```

```
not (4 != 7)
```

```
False
```

```
not( (3 > 3 ) and ( 6 < 3))
```

```
True
```

```
not( (3 > 3 ) or ( 6 < 3))
```

```
True
```

```
not( (3 >= 3 ) or ( 6 < 3))
```

```
False
```

Membership Operator

```
"a" in "Aditya"
```

```
True
```

```
"b" in "Aditya"
```

```
False
```

```
"c" not in "Aditya"
```

```
True
```

```
"D" in "Aditya"
```

```
False
```

```
"d" in "Aditya"
```

```
True
```

Quizes

```
1. What will be the output?

print(10 + 5)

A. 105

B. 15

C. 50

D. Error

Ans: B
```

```
2. What will be the output?

print(20 - 8)

A. 12

B. 28

C. -12
```

D. Error

Ans: A

3. What will be the output?

```
print(6 * 4)
```

A. 10

B. 24

C. 64

D. Error

Ans: B

4. What will be the output?

```
print(15 / 3)
```

A. 5

B. 5.0

C. 4.0

D. Error

Ans:

Ans: B

5. print(17%5) --> 2

6. print(4 ** 2) --> 16

7. print(20//3)

```
print(20//3)
```

```
6
```

8.
Predict the output.

```
x = 10

x += 5

print(x)
```

Ans: 15

9. Predict the output.

```
salary = 50000

salary -= 10000

print(salary)
```

A. 60000

B. 50000

C. 40000

D. Error

Ans: C

10. What will be the output?

```
marks = 40

print(marks >= 40)
```

A. True

B. False

C. 40

D. Error

Ans: A

11. What will be the output?

```
print(True and False)
```

A. True

B. False

C. None

D. Error

Ans: B

HW

What will be the output?

```
x = 5

x += 3

x *= 2

print(x)
```

A. 8

B. 11

C. 16

D. 13

Predict the output:

```
age = 20

print(age >= 18 and age < 60)
```

```
print(age >= 18 and age <= 60)

A. True

B. False

C. 20

D. Error
```